Introduction:

Around the world human activities which includes mining, deforestation, hunting, agriculture, logging and more, have had a big impact in the Madagascar forests ecosystem. The indri species found in the Makira Forest in north-eastern Madagascar which covers 371,217 ha of lowland and mid altitude rainforest (Golden, 2009). According to the IUCN\SSC Primate specialist Group of Madagascar: an action plans for their conservation, it is stated that the Indri species has been given the Highest priority rating for being one of the endangered species in Madagascar (Quinn and Wilson, 2002). All the indri species are listed in the appendix 1 of 1973 conservation on International Trade in Endangered species (Quinn and Wilson, 2002). Madagascar has one of the most rapidly growing populations of countries that possess rainforest therefore land use of Marika Forest is turned to human settlement, and wildlife habitats are depleted (Golden, 2009). The intriguing thing is that the Indri species reaches sexual maturity between 7 and 9 years of age which means the female reproduce between 2 and 3 years after a period of gestation of 120-150 days (Quinn and Wilson, 2002). Humans thereby affects the mating process of the indris through them activities which cause a reduction to produce the next generation. The main diet for the Indris b species leaves, fruit, flowers, buds from any trees (Quinn and Wilson, 2002), Indri lemurs live mostly in trees within troops of 6-8 members. In the last 10 years the indri population has been declining at a very fast rate with only 10 000 of these species left in the world. Through habitat destruction their diet is affected which contribute to their dead. Lastly, actions would need to be taken to eliminate or reduce threats and describe additional objectives and actions to ensure the long-term conservation of the Indri population in the Makira Forest and their habitats while also improving human livelihoods (Mittermeier et al., 1992).



Figure 1: pictures of all the factors affecting the indri population.



Figure2: flow chart of the document

Project scope: (thematic)- includes areas in Madagascar where the indri lemur population are viable and can be restored.

Vision statement:

We seek to prevent the extinction of the largest lemur in the Makira Forest north-eastern of Madagascar, by implementing habitat restoration which includes encouraging afforestation, reducing illegal and over hunting of these species as well as to raise awareness for the conservation of these species in the next coming decade, which will reverse the currently proposed extinction.

Targets and KEA'S:

Indicator ratings

TARGET	CATEGORY	KEA	INDICATOR	POOR	FAIR	GOOD	VERY GOOD
Indri lemur	size	Troop size	# Adult and pups lemurs	0-1	2	3-5	≥6
	current	status				5	
desired	future	status					≥6
Indri	abundance	Abundance of the indri lemurs in Madagascar forests	# Of the indri lemurs in Madagascar forests	≤10 000	10 000-15 000	›15 000-20 000	30 000
	current	status		10 00			
desired	future	status				16 000	
Controlled hunting	Management	Abundance of local hunter	Percentage Of families still dependent on lemurs for food source	85 %-100%	84%-60%	60-40%	≤39%
	current	status			≤84%		
desired	future	status				60-40%	
Habitat conservation	condition	Abundance of tree species in Madagascar	# Of new adult trees	0-30	40-50	60-80	≤90
	current	status			≥40		
desired	future	status				75	
Awareness of the need	Education	Social media coverage	# Of participants	≥ 100	150-200	220-400	≥500

for conservation of the indri		and all-round spread of the conservation plan	in conservation programs, success in lemur populations				
	current	status		›100			
desired	future	status				≥220	
Reproductive success	Mating partners	Abundance of male and female lemurs in each troop	Number of new offspring per year	0-200	300-400	500-900	≥1000

Current status	≥100			
Desired future status			›500	

Threats:

Direct threats to the	**scope**	**severity**	**Irreversibility**
Hunting/poaching	High	High	medium
Diseases	high	high	high
Shooting	high	high	low
Trafficking	high	high	low

Direct threats to The habitat	**scope**	**severity**	**Irreversibility**
deforestation	high	High	medium
Habitat destruction	high	high	high
Mining	medium	high	High
Forest fires	high	High	high
Invasive species	low	medium	medium

Indirect threats	**scope**	**severity**	**irreversibility**
poverty	high	Medium	medium
Lack of knowledge and awareness of the conservation problem	low	low	low
Lack of veterinary programs	high	high	low
LACK OF CULTURAL GROUP MANAGEMENT	low	medium	Medium
LACK OF EMPLOYMENT OPPORTUNITIES for local people of Madagascar	low	low	low
Mining of treethanol made from tree in the Madagascar forests	high	medium	medium

Strategies:

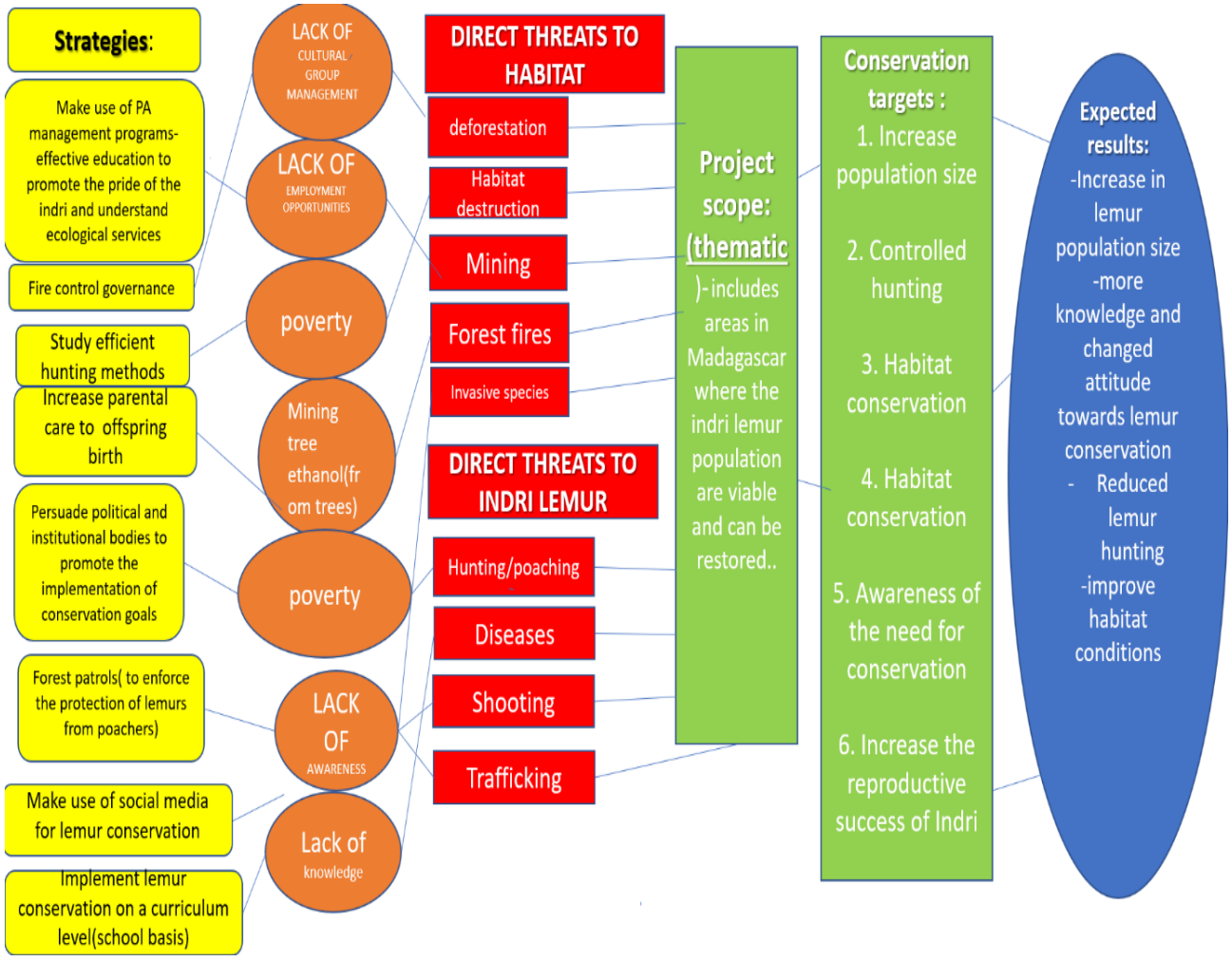
colour coded rating key:

Good strategy	
Bad strategy	
Okay	

strategy	Potential impact	feasibility
Make use of PA management programs- effective education to promote the pride of the indri and understand ecological services	- Raise awareness of conservation needs -	- Highly accomplishable
Fire control governance	- Reduce frequent occurrence of forest fires	- Medium, - Due the recent unpredictable climatic conditions., implementing this strategy will be difficult if not impossible
Persuade political and institutional bodies to promote the implementation of conservation goals	Raise money to fund conservation project	- High
Isolate species in environments that encourage reproductive success	- Increase population size and mating partners - more successful reproduction - more offspring -	- Medium, we can't confirm reproductive success, there are hybrids may be inviable and due to the complexity of mating calls, mating may not occur
Increase parental care to offspring birth	- Ensure longer life span -	- Very high, maternal care is verified
Forest patrols(to enforce the protection of lemurs from poachers)	- Reduce hunting and poaching - Control deforestation - Employment opportunities	- Low, funds may be limited

Isolate mother to protect her from predators(humans)	- Ensure longer life span - Infant has higher chance of survival - Protection from predators	- Low , most troops are made of two adults a male and female. To take the female away would change the dynamic of the troop
Arrest illegal hunters	- Reduced number of hunters - Increase population size	- Low, the ethics and merit behind arresting them promotes injustice
Make use of social media for lemur conservation	- Raise awareness for conservation needs - Raise funds for conservation project	-high, not everyone has social media access
Implement lemur conservation on a curriculum level (school basis)	- Raise awareness for conservation needs - Youth becomes better adapted to strategic implementation of conservation needs	- Very high, it is easier to teach young kids than it is to teach adults
Relocate lemurs to a safer environment	- Increase population size - Increased chances of lemur survival	- Low, it will change the ecological niche - There isn't enough land space to accommodate lemurs
Study efficient hunting methods	- Control hunting - Reduce illegal hunting	- High, we can't confirm the willingness of communities to work towards conservation targets
Threat abandonment strategy	- Reduce hunting - Raise funds from the money obtained from illegal poachers	- High

Conceptual model:



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